Control Structures: Loops and Conditionals

In this lecture, we will explore control structures in Python, focusing specifically on loops and conditional statements. These structures are essential for controlling the flow of your programs, allowing you to execute certain blocks of code based on conditions or to repeat actions multiple times. By the end of this lecture, you will be able to implement for loops, while loops, and if/else statements in your Python programs.

Introduction to Control Structures

Control structures are fundamental components of programming that dictate the order in which instructions are executed. In Python, the two primary types of control structures we will discuss are loops and conditionals.

Loops

Loops allow you to execute a block of code multiple times. Python provides two main types of loops: for loops and while loops.

For Loops

A for loop is used to iterate over a sequence (like a list, tuple, or string) or a range of numbers. The syntax for a for loop is as follows:

```
for variableName in groupOfValues:

statements
```

Example:

```python
for x in range(1, 6):

print(x, "squared is", x * x)
```

Output:

```
1 squared is 1

2 squared is 4

3 squared is 9

4 squared is 16

5 squared is 25
```

In this example, the loop iterates over the numbers 1 through 5, calculating and printing the square of each number.

Range Function

The `range()` function generates a sequence of numbers. It can take one, two, or three arguments:

- `range(stop)` - generates numbers from 0 to stop (exclusive).

- `range(start, stop)` - generates numbers from start to stop (exclusive).

- `range(start, stop, step)` - generates numbers from start to stop (exclusive) with a specified step.

Example:

```python
for x in range(5, 0, -1):
print(x)
print("Blastoff!")
```

Output:

```
5
4
3
2
1
Blastoff!
```

While Loops

A while loop continues to execute as long as a specified condition is true. The syntax is:

```
while condition:
statements
```

Example:

```python
number = 1
while number < 200:
print(number)
number = number * 2
```

Output:

```
1

2

4

8

16

32

64

128
```

In this example, the loop prints the value of `number` and doubles it until it reaches 200.

Conditionals

Conditional statements allow you to execute certain blocks of code based on whether a condition is true or false. The primary conditional statements in Python are `if`, `elif`, and `else`.

If Statement

The `if` statement executes a block of code if a specified condition is true. The syntax is:

```
if condition:

statements
```

Example:

```python
gpa = 3.4

if gpa > 2.0:

print("Your application is accepted.")
```

If/Else Statement

The `if/else` statement allows you to execute one block of code if the condition is true and another block if it is false. The syntax is:

```
if condition:

statements

else:

statements
```

```

**Example:**

```python
gpa = 1.4
if gpa > 2.0:
print("Welcome to Mars University!")
else:
print("Your application is denied.")
```

Elif Statement

You can chain multiple conditions using `elif` (else if). The syntax is:

```
if condition:
statements
elif condition:
statements
else:
statements
```

Example:

```python
x = 30
if x <= 15:
y = x + 15
elif x <= 30:
y = x + 30
else:
y = x
print('y =', y)
```

Conclusion

In this lecture, we covered the essential control structures in Python: loops and conditionals. We learned how to use for loops to iterate over sequences, while loops for indefinite repetition, and conditional statements to control the flow of execution based on conditions. Mastering these
```

concepts is crucial for writing effective and efficient Python programs. In the next lecture, we will explore functions and how they can help us organize our code better.

Quiz Questions

1. What are the two primary types of control structures discussed in the lecture?

    - Loops and Functions
    - Loops and Conditionals
    - Functions and Classes
    - Variables and Data Types

Correct: Loops and Conditionals

2. What is the purpose of a for loop in Python?

    - To execute a block of code once
    - To iterate over a sequence or range of numbers
    - To define a function
    - To create a conditional statement

Correct: To iterate over a sequence or range of numbers

3. Which statement is used to execute a block of code if a specified condition is false?

    - if
    - elif
    - else
    - for

Correct: else